IT ROOTS
EMPOWERING MINDS

#Empowering Minds

# Rooted | Recruited

ROOTED IN SKILLS, RECRUITED FOR SUCCESS.

# Data Analytics with Python

TRANSFORMING POTENTIAL INTO PROFESSIONALS.

www.itroots.co.in

# About ITROOTS

At ITROOTS, we believe in transforming potential into professional excellence.
With over two years of proven success in Online training, ITROOTS now brings its industry-focused expertise OFFLINE , empowering learners across India to become job-ready professionals.

**Our mission** is to equip every learner with industry-ready skills, hands-on experience, and career confidence through AI-integrated, project-based learning.

# Why Choose ITROOTS?

Choosing the right training partner can define your career and at ITROOTS, we turn your goals into guaranteed outcomes.

### Why learners and employers trust us:

**100% Placement Guarantee:** We ensure every learner achieves their career goal.

**Refund Guarantee:** If we don't deliver as promised, your investment is safe.

**Post-Placement Support:** Continued guidance even after you're hired.

**Lifetime Career Assistance:** Ongoing mentorship and growth support

**Project-Oriented, AI-Integrated Training:** Learn through live, industry-level projects.

**Expert Mentorship:** Trainers with 10+ years of industry experience

**Daily 5–6 Hours of Practical Training:** Focused, hands-on sessions.

**Regular Assessments & Mock Interviews:** Build confidence before stepping into interviews.

*At ITROOTS, you don't just learn, you build, apply, and grow with every step.*

# Program Overview

The Data Analytics with Python is a 6-month intensive course crafted to make you an industry-ready analytics expert.

You'll gain hands-on experience with Excel, SQL, Power BI, Python, and AI-based analytics tools to turn data into powerful business insights and strategic decisions.

**Duration:** 6 Months **Mode :** Online / Offline

**Ideal for:** Graduates, Job Seekers, and Working Professionals

# The Learning Journey

Every learner at ITROOTS follows a path that ensures real progress and measurable results:

**Enroll & Set Goals**
Begin your transformation with a clear career roadmap.

**Learn Intensively**
Master Data Analytics through focused, daily sessions.

**Build & Innovate**
Work on real, AI-integrated projects that sharpen your skills.

**Test & Improve**
Strengthen your performance with assessments and mock interviews.

**Get Placed**
Land your dream job with our 100% placement guarantee.

**Keep Growing**
Enjoy lifetime mentorship and career guidance - always.

# Syllabus

## Python Programming Fundamentals

Introduction to Python | Python Installation & Environment Setup | Understanding Syntax & Indentation | Variables & Data Types | Operators & Expressions | Conditional Statements – if, elif, else | Looping Constructs – for, while, nested loops
Hands-On Practice + Weekly Test + Q&A

## Data Structures & Core Python Concepts

Data Structures & Core Python ConceptszLists – Creation, Indexing, Slicing | Tuples – Immutable Sequences | Sets – Operations & Use Cases | Dictionaries – Key-Value Pairs | Advanced List & Dictionary Comprehensions
Hands-On Coding + Practice Assignments + Q&A

## Functions & Object-Oriented Programming (OOP)

Defining Functions – Parameters & Arguments | Default, Keyword & Lambda Functions | Variable Scope – Local & Global | Classes & Objects | Constructor (init) | Inheritance – Single & Multiple | Polymorphism | Encapsulation | Practical OOP Implementation
OOP Assignments + Coding Challenges + Q&A

## File Handling, Exceptions & Python Ecosystem

File Handling – Text, CSV, JSON | File Operations – Open, Read, Write, Close | Exception Handling – try, except, finally | Custom Exceptions | Built-in Modules – math, datetime, os | Creating Custom Modules | Packages & Virtual Environments (venv)
Hands-On Practice + Weekly Test + Q&A

## Web Development with Django

Django Basics & Web Development Concepts | HTTP Requests & Responses | Django Project & App Structure | Models & ORM – Migrations, CRUD Operations | QuerySets | Views & URLs – Dynamic Routing | Templates & Static Files | Forms & Validation | Authentication & Admin | Django REST Framework – APIs & Postman | Git & Deployment Basics
Mini Project + Practical Labs + Q&A

# Syllabus

## Excel for Data Analysts

Excel Basics | Tables & Ranges | Conditional Formatting | Sorting & Filtering | Charts & Graphs | Pivot Tables | Lookup Functions – VLOOKUP, HLOOKUP, XLOOKUP | Data Cleaning | What-If Analysis | Macros & Productivity Shortcuts
Practical Exercises + Business Use Cases + Q&A

## SQL for Data Analytics

Relational Database Concepts | MySQL / PostgreSQL Basics | SELECT, INSERT, UPDATE, DELETE | Filtering & Sorting | Joins – Inner, Left, Right, Full | GROUP BY & HAVING | Aggregate Functions | Subqueries | Views & Indexing | Data Normalization
Hands-On Queries + Practice Sets + Q&A

## Python for Data Analysis

Python Refresher for Analytics | Jupyter Notebooks | Working with CSV & JSON | NumPy – Arrays, Indexing, Slicing | Pandas – DataFrames & Series | Data Cleaning & Transformation | Handling Nulls, Outliers & Duplicates | Date-Time Operations
Hands-On Labs + Weekly Test + Q&A

## Data Visualization & Statistics

Data Visualization – Matplotlib, Seaborn, Plotly (Intro) | Charts – Line, Bar, Histogram, Boxplot | Heatmaps & Correlation | Excel vs Python Visualization | Descriptive Statistics | Probability Basics | Distributions | Hypothesis Testing – z-test, t-test | Outlier Detection
Visualization Tasks + Statistical Analysis + Q&A

## Power BI & Tableau (BI Tools)

Power BI – Data Loading, Power Query, Visualizations, Filters, DAX Functions | Interactive Dashboards | Sharing & Publishing Reports | Tableau – Data Connections, Visualizations, Dashboards, Stories | Tableau Public Publishing
Dashboard Projects + Case Studies + Q&A

# Syllabus

## AI & Machine Learning with Python

Introduction to AI, ML & DL | Real-World Use Cases | ML Workflow | Data Preprocessing | Train-Test Split | Scikit-Learn Overview | Supervised Learning – Linear Regression, Logistic Regression, KNN | Evaluation Metrics | Unsupervised Learning – K-Means Clustering | PCA (Intro)
Mini ML Project + Hands-On Practice + Q&A

## Capstone Project & Career Readiness

Capstone Project – Real-World Dataset | Data Cleaning, EDA, Model Building & Evaluation | End-to-End Report & Presentation | Console-Based Python Project | Resume Building | Mock Interviews | Final Assessment | Full Revision & Q&A Wrap-Up
Industry-Oriented Capstone + Placement Preparation

## Aptitude, Communication & Career Readiness

Quantitative Aptitude – Number Systems, Simplification, Ratio & Proportion, Percentages, Averages, Profit & Loss, Simple & Compound Interest, Time & Work, Time-Speed-Distance, Data Interpretation | Logical Reasoning – Series (Numbers & Alphabets), Coding-Decoding, Blood Relations, Direction Sense, Seating Arrangement, Puzzles, Syllogism, Venn Diagrams, Ranking & Ordering, Clock & Calendar | Verbal Ability – Vocabulary Building, Grammar (Tenses, Articles, Prepositions), Sentence Framing, Error Spotting, Reading Comprehension, Daily Use Expressions | Communication & Soft Skills – Verbal & Non-Verbal Communication, Body Language, Active Listening, Effective Speaking, Group Discussion Practice, Presentation Skills, Public Speaking, Overcoming Stage Fear | Career & Interview Preparation – Resume Writing, Elevator Pitch, HR Interview Questions, STAR Technique, Mock Interviews, Workplace Etiquette, Time Management & Goal Setting
Weekly Tests + Practice Exercises + Mock Interviews + Final Assessment + Q&A

# Data Analytics with Python Syllabus

## Module 1

### Introduction to programming languages

- Need of Database
- DBMS Models
- Normalization •
- SQL Constraints
- SQL Commands(DML,DDL,DQL,TCL,DCL)
- Clauses
- Functions
- Joins
- Indexes ,Views
- PLSQL block structure
- Difference between Stored Procedure,Functions
- Exercises

## Module 2

### Excel

- Master Data Cleaning Techniques
- Visualize Data with Excel Charts
- Efficient Sub-totaling and Analysis

### Advanced Excel

- Pivot Tables forData Summarization: Understand how to summarize data efficiently.
- Data Analysis and Visualization: Deepen your data analysis and visualization skills.

## Module 3

### DWH & ETL

- Database vsData Warehouse:
- Key Differences • Data Warehouse Architecture, Concepts and Components
- Types of Schemas (Star, Snowflake)
- Data Warehouse vs Data Mart
- Data Lake vs Data Warehouse: What's the Difference?
- ETL (Extract, Transform, and Load) Process
- What is Business Intelligence? Definition & Example

## Module 4

### TABLEAU

- Introduction to Tableau
- Data Connectors
- Data Model Basic Visualizations
- Filters
- Groups
- Set
- Types of Calculations (Basic, Table, LOD)
- Date
- Functions Table Calculations First()
- Last()
- Index()
- Data Blending Advance Visualizations
- Gant Chart, Histogram, Donut Chart, Lollipop
- Chart) Dashboards ProjectsZZ

### Power BI

- Difference between PowerBI and Tableau
- DataSources
- Key Terms
- Installation
- Filters
- DAX Function
- Visualization
- Projects

# Projects

## RetailPro: Smart Inventory Manager

Optimize inventory levels and reduce waste across retail chains by analyzing sales trends, restock cycles, and seasonal demand.

**Technology: Data Analytics**

## FinScope: Business Expense Tracker

Track, visualize, and optimize company-wide expenses and revenues for better budgeting and forecasting.

**Technology: Business Analytics**

## MetroCommute: City Transport Analysis

Evaluate public transport performance using ticket data and route occupancy trends to recommend route optimization.

**Technology: Data Analytics**

## ShopTrend: Online Store Insights Dashboard

Build an interactive dashboard that showcases online product trends, customer preferences, and category performance.

**Technology: .NET**

## QuizBash: Interactive Coding Quiz Platform

A real-time coding quiz platform with scoring and feedback, designed to test job-ready skills.

**Technology: Java**

## WeatherSnap: Forecast Dashboard System

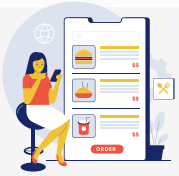
Build a system that fetches and displays real-time and historical weather data with filtering options.

**Technology: Java**

## DineSmart: Restaurant Order Manager

Create a complete restaurant order management and billing solution with admin and kitchen modules.

**Technology: .NET**



# Our Role in Your Success

At ITROOTS, we're not just trainers, we're your career partners. Our goal is to help you build practical skills, gain real-world exposure, and step confidently into your professional journey. With dedicated mentorship, industry tools, and continuous support, we make sure you grow, improve, and stand out.

## 1. Skill Development & Practical Learning

We train you in the most in-demand Data Science tools and technologies.
You'll master:

- Python & Data Analysis | NumPy & Pandas
- Data Visualization – Matplotlib, Seaborn
- SQL & Databases
- Machine Learning fundamentals
- Power BI & Tableau
- Hands-on training with real datasets and projects to make you job-ready.

## 2. Continuous Career Support

We don't just train you; we prepare you to thrive.
Our dedicated career team provides:

- Resume & LinkedIn optimization
- Technical & HR mock interviews
- Personalized career counselling
- Access to exclusive job networks and hiring partners
- Lifetime mentorship for continuous growth and upskilling

## 3. Guaranteed Placement

Your success is our goal.
Through our Hire → Train → Deploy model, we ensure every learner is:

- Trained with AI-integrated, project-based sessions
- Evaluated through tests and mock interviews
- Deployed into top IT companies with complete career readiness

With our 100% Placement Guarantee and Refund Assurance, your journey with ITROOTS is result-driven, transparent, and focused on success.

# Alumni Success Stories

ITROOTS is one of the best institutes for training and placement. The trainers are experienced, supportive, and always approachable. I completed my Full Stack Java Certification from ITROOTS and got placed as a Software Developer. Grateful to the entire ITROOTS team for their constant guidance.
~ **Priya Wagh**

Joining ITROOTS was the best decision for my IT career. In just six months, I learned Frontend, Backend, and Database technologies and got placed within a week after completing the course. The placement team is truly dedicated and helpful throughout the process. **~ Harsh Patil**

ITROOTS is an excellent place for learning and placement. The team is very supportive and helps you build knowledge from the basics, even if you come from a non-IT background. Thank you, ITROOTS, for helping me start my career with confidence. **- Mayur Aware**

250+ Success stories so far and counting
visit our website for more success stories and get inspired

accenture | bitwise | Capgemini | Clarion | AFOUR TECHNOLOGIES | Technosoft CORPORATION

Survik SOFTWARE Innovative Resourcing | TATA TATA TECHNOLOGIES | Allscripts | AMSOFT GROUP | tcs TATA CONSULTANCY SERVICES | cognizant

SYNTEL Consider IT Done | Synechron | IBM | Tech Mahindra | calsoft | HCL

and many more....

# Our Commitment

***"Rooted in Skills. Recruited for Success."***

At ITROOTS, we promise outcomes, not just education. With 100% Placement Guarantee, Refund Assurance, and Project-Oriented Training, every learner becomes a professional with confidence and clarity. We prepare you for industry success with real skills, real projects, and lifelong support.

***Your growth doesn't stop at placement, it starts there.***

# Contact Us

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